

Hailey Garcia

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Department of Mathematics, Louisiana State University, Baton Rouge, LA 70803

Education

Ph.D. Mathematics, Louisiana State University	Fall 2025–present
<i>Advisor: Christin Bibby</i>	
B.S. Mathematics, Louisiana State University	Fall 2022–Spring 2025

Papers and Preprints

G. Agrahari, C. Bibby, S. Boros, H. Garcia, F. Heidercheidt, Z. Wang. *On some structural properties of graphs with non-negative resistance curvature*. arXiv:2607.13169 (2026). [arXiv:2607.13169](https://arxiv.org/abs/2607.13169).

M. Fairchild, H. J. Garcia, J. Murphy, H. Percle. *On the non-orientable 4-genus of torus knots with braid index 5 or 6*. *Journal of Knot Theory and Its Ramifications* **35** (2026), No. 7, 2650001. <https://doi.org/10.1142/S021821652650001X>.

Talks

Noncontractive Function Systems	Spring 2023
<i>6th TX-LA Mathematics Undergraduate Conference</i>	
On The “Kelly Sequence”	Spring 2023
<i>6th TX-LA Mathematics Undergraduate Conference</i>	

Teaching

MATH 1431, Business Calculus — Lab Leader/TA, LSU	Spring 2026
<i>Sections 7, 11, 19</i>	
MATH 1431, Business Calculus — Lab Leader/TA, LSU	Fall 2025
<i>Sections 7, 8, 9, 11, 25, 28</i>	
Honors Calculus 1 — Primary Instructor, Arden Cahill Academy	Fall 2023–Spring 2024

Service

TA, 2026 LSU RTG Summer Undergraduate Workshop	Summer 2026
Webmaster, RTG@LSU	Spring 2026–present
Manager, LSU Math 3D Print Lab	Spring 2026–present
Officer, LSU AWM Student Chapter	Fall 2025–present
<i>President (Fall 2025–Spring 2026); Secretary (Spring 2026–present);</i>	
<i>Webmaster (Fall 2025–present)</i>	

Mentor, Vertically Integrated Research Courses, LSU

Fall 2025–present

Awards

Pasquale Porcelli Graduate Student Academic Excellence Award

Spring 2026

Robert and Betti Giles Senior Mathematics Award

Spring 2025

Peg and Tom Madden Mathematics Research Scholarship

Spring 2024

Porcelli Junior Scholarship

Spring 2023